

TrES-3b

R-0573

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240323 · created 2026-07-24T16:34:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460392.9113 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.057
Transit depth (Rp/Rs)^2	1.16 +/- 1.23 [%]
Orbital Inclination (inc)	80.53 +/- 2.82
Airmass coefficient 1 (a1)	0.39 +/- 0.21
Airmass coefficient 2 (a2)	0.55 +/- 0.87
Scatter in the residuals of the lightcurve fit is	46.04 %
Best Comparison Star	#3 - [321, 396]
Optimal Aperture	1.7
Optimal Annulus	7.0
Transit Duration (day)	0.03 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.108 $\pm$ 0.057 vs 0.16309 (deviation 0.97 σ)
Duration fit vs published	0.03 d vs 0.0567 d (deviation 0.99 σ)
Mid-time O-C	3.4 min
Depth SNR	1.9
Residual scatter	46.04 %

Light curve

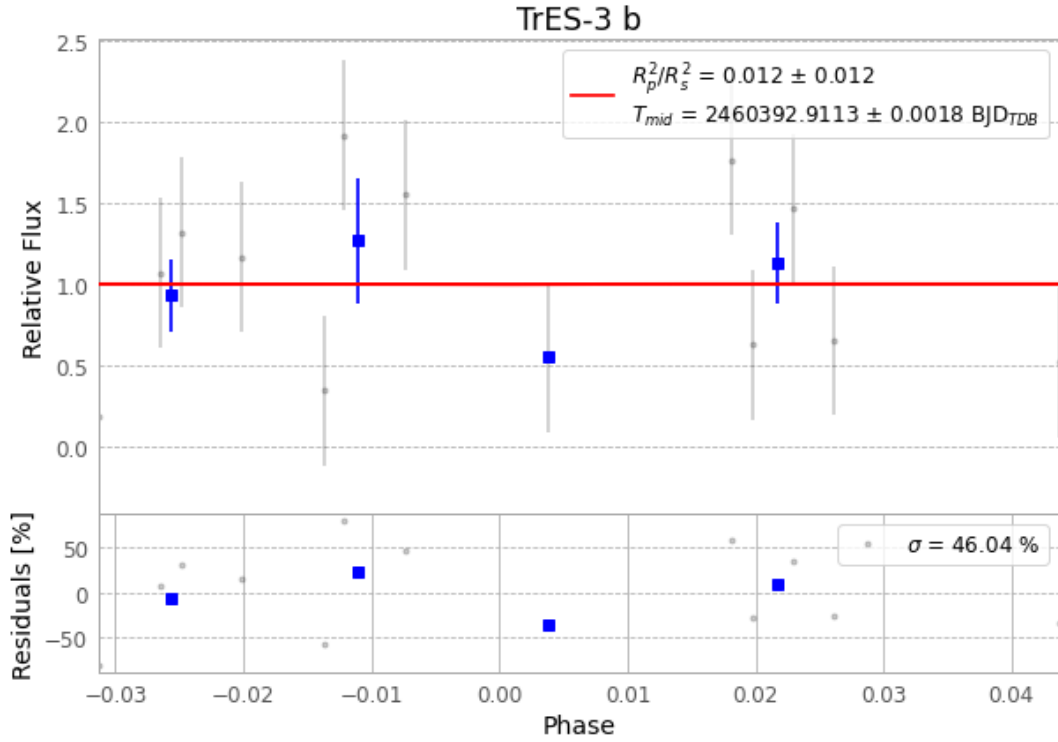


Figure 1 — FinalLightCurve_TrES-3 b_2024-03-23.png (SHA-256 61b07e511054a9ca...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [402, 147], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8e3c77c7386e02dc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

61 FITS frames (40 MB), TRES-3240323083015.FITS → TRES-3240323113015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240323.log (3abc9f9158ba...), FinalLightCurve_TrES-3 b_2024-03-23.png (61b07e511054...), FinalLightCurve_TrES-3 b_2024-03-23.csv (1fc6d23af5e7...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 16:34Z.